

Deploying Static Website using S3 Bucket

Objective

The objective of this project is to demonstrate how a static website can be hosted using Amazon S3, a cloud storage service provided by Amazon Web Services. This project helps in understanding how cloud platforms can be used to deploy and manage website content.

Description

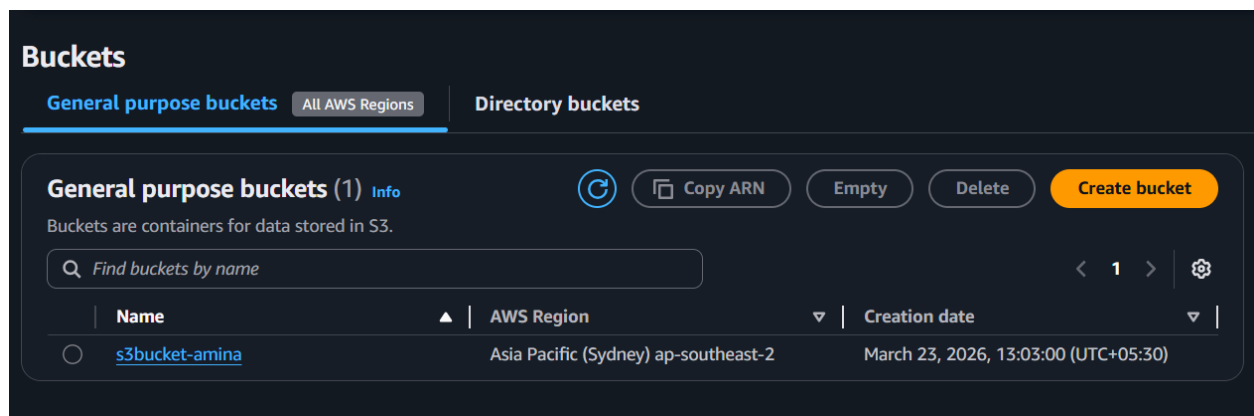
Static websites contain fixed content such as HTML pages, CSS files, and images. In this project, the website files are stored in an Amazon S3 bucket and made publicly accessible through the internet. By enabling static website hosting in the S3 bucket, users can access the website using a generated endpoint URL without the need for a traditional web server.

Tools Required

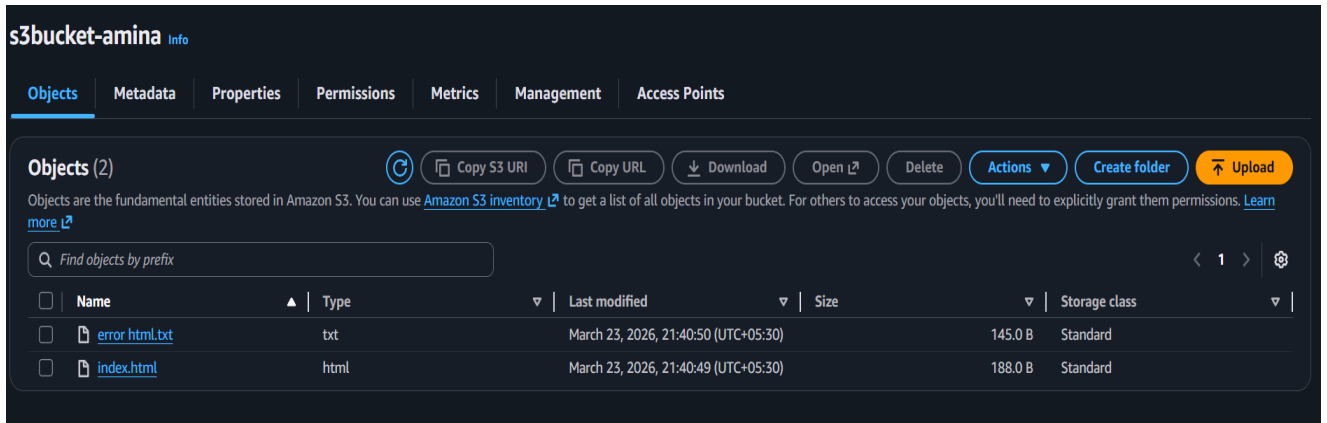
- AWS Free Tier Account
- Web Browser

Implementation

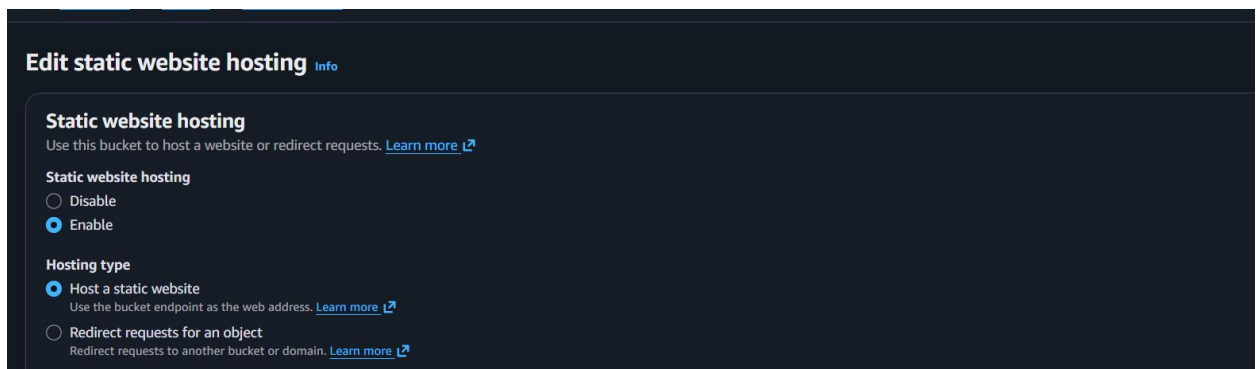
1. Logged into the AWS Management Console.
2. Opened the Amazon S3 service.
3. Created a new S3 bucket with a unique name.



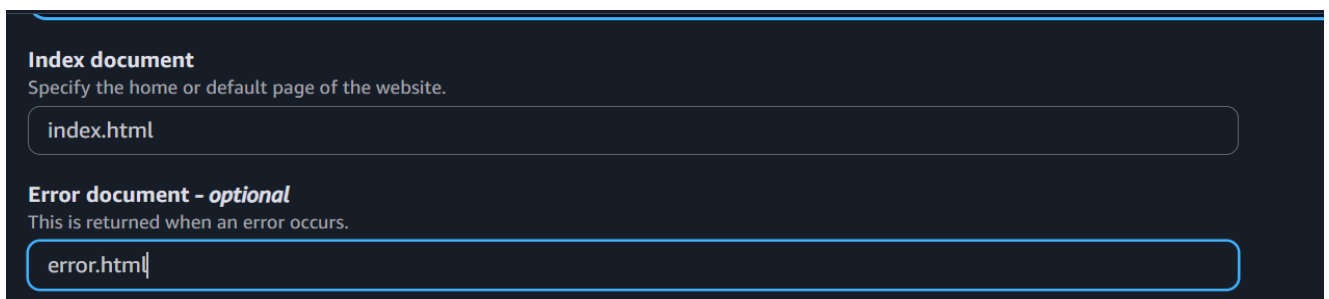
4. Uploaded the website files such as index.html and error.html into the bucket.



5. Enabled Static Website Hosting in the bucket properties.



6. Specified the following documents:
 - Index document: index.html
 - Error document: error.html



7. Generated and applied a bucket policy to allow public access to the website files.

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through *new* access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through *any* access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through *new* public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through *any* public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Access control list (ACL)

Grant basic read/write permissions to other AWS accounts. [Learn more](#)

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID:  a871782ad4e4235db5e5f9952c084fbd8def8addc1cf895999efc6c1f9035fd	☒ List ☒ Write	☒ Read ☒ Write
Everyone (public access) Group:  http://acs.amazonaws.com/groups/global/AllUsers	☒  List ☐ Write	☒  Read ☐ Write
Authenticated users group (anyone with an AWS account) Group:  http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ List ☐ Write	☐ Read ☐ Write
S3 log delivery group Group:  http://acs.amazonaws.com/groups/s3/LogDelivery	☐ List ☐ Write	☐ Read ☐ Write

Bucket policy

[Edit](#)[Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::s3bucket-amina/*"
    }
  ]
}
```

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- Used the generated S3 website endpoint URL to access the hosted website.



- Tested and verified that the website loads correctly.



Welcome to My Website

This website is hosted using AWS S3.

Error Page

Page not found!

Result

The static website was successfully deployed and hosted using Amazon S3 provided by Amazon Web Services. The website files, including index.html and error.html, were uploaded to the S3 bucket and static website hosting was enabled. The website was accessed through the generated S3 website endpoint URL in a web browser.